Auditable AI Agents: An Immutable On-Chain Audit Trail for LLM Agent Actions

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***Abstract*—Autonomous agents built on large language models (LLMs) call tools and act on behalf of users, yet their records are usually ordinary log files that anyone with server access can edit or delete without leaving evidence. This paper presents a tamper-evident audit trail for LLM agent actions. Each step, namely the prompt, model response, tool call, and tool result, is written to a hash-chained log, and batches of entries are folded into Merkle roots anchored in a smart contract, so only 32-byte hashes, never raw logs, are stored on-chain. A verifier recomputes hashes from the stored log, compares them with the anchored roots, and locates the first altered entry or batch. The prototype, implemented in Python and Solidity on a local Ethereum development chain, was exercised with a local open-weight model. In randomized tests, seven classes of tampering (edits to payloads, timestamps and hashes, deletion, insertion, reordering, and truncation) were detected in all 2,100 trials, with no false alarms on 300 untouched logs, and a consistently rewritten log, which defeats a hash chain alone, was caught by the on-chain roots. Python-generated Merkle proofs verified on-chain for trees of up to 100 entries, and anchoring cost roughly 80,000 to 93,000 gas per batch in a first run. Gas-versus-batch-size, overhead, and scaling studies are under way. The trail proves a log was not altered after anchoring, not that agent outputs were correct.**

***Index Terms*—*LLM agents, audit trail, tamper-evident logging, Merkle tree, blockchain anchoring, smart contracts, AI accountability.*